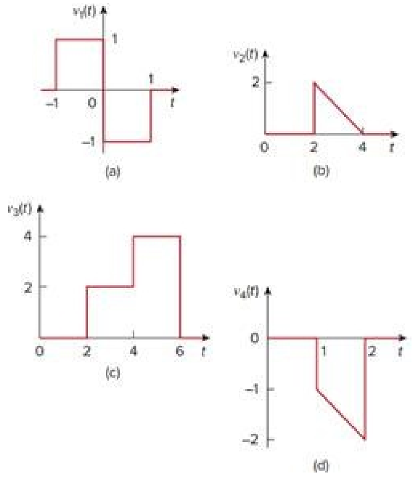
Chapter 7, Problem 26P

Problem

Express the signals in Fig. 7.104 in terms of singularity functions.

Fig. 7.104:



Step-by-step solution

Step 1 of 4

(a)

Refer to the Figure 7.104(a) in the text book.

Draw the following diagram.

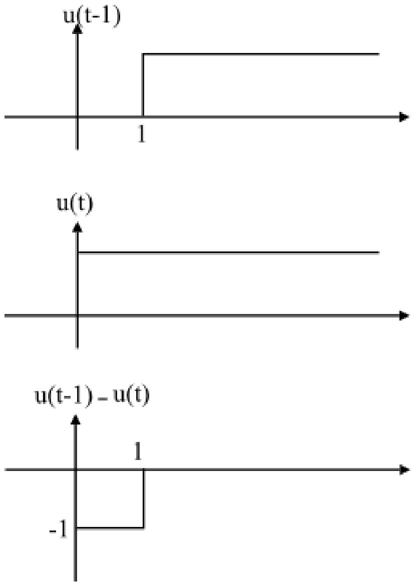


Figure 1

Draw the following diagram.

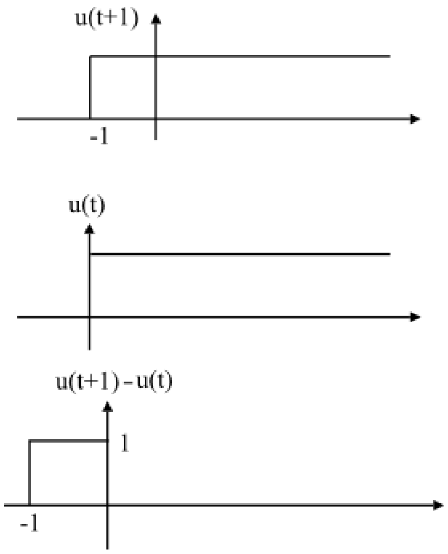


Figure 2

Step 2 of 4

Add Figure (1) and (2) to obtain Figure 7.104(a).

$$\begin{aligned} V_i(t) &= \left[u(t+1)-u(t)\right]+\left[u(t-1)-u(t)\right] \\ &= u(t+1)-2u(t)+u(t-1) \end{aligned}$$

Therefore, the expression of $V_i(t)$ is $\boxed{u(t+1)-2u(t)+u(t-1)}$.

Step 3 of 4

(b)

Refer to the Figure 7.104(b) in the text book.

Determine the equation of line between the points $(2,2)$ and $(4,0)$

$$\begin{aligned}
V_2(t)-2&=\left(\frac{0-2}{4-2}\right)(t-2)\\
V_2(t)-2&=-1(t-2)\\
V_2(t)&=-t+4
\end{aligned}$$

But the Figure is lies between 2 and 4. Hence, multiply the expression with $\left[u(t-2)-u(t-4)\right]$

$$\begin{aligned}
V_2(t)&=(-t+4)\left[u(t-2)-u(t-4)\right]\\
&=(4-t)u(t-2)-(4-t)u(t-4)\\
&=-(t-2-2)u(t-2)+(t-4)u(t-4)\\
&=-(t-2)u(t-2)+2u(t-2)+(t-4)u(t-4) \qquad \left[\text{Since } tu(t)=r(t)\right]\\
&=-r(t-2)+2u(t-2)+r(t-4)
\end{aligned}$$

Therefore, the expression of $V_2(t)$ is $\boxed{-r(t-2)+2u(t-2)+r(t-4)}$.

(c)

Refer to the Figure 7.104(c) in the text book.

The Figure ranges from in the interval 2 to 4 with amplitude 2.

$$V_3(t)'=2\left[u(t-2)-u(t-4)\right]$$

The Figure ranges from in the interval 4 to 6 with amplitude 4.

$$V_3(t)''=4\left[u(t-4)-u(t-6)\right]$$

Write the expression of $V_3(t)$.

$$\begin{aligned}
V_3(t)&=V_3(t)'+V_3(t)''\\
&=2\left[u(t-2)-u(t-4)\right]+4\left[u(t-4)-u(t-6)\right]\\
&=2u(t-2)-2u(t-4)+4u(t-4)-4u(t-6)\\
&=2u(t-2)+2u(t-4)-4u(t-6)
\end{aligned}$$

Therefore, the expression of $V_3(t)$ is $\boxed{2u(t-2)+2u(t-4)-4u(t-6)}$.

Step 4 of 4

(d)

Refer to the Figure 7.104(d) in the text book.

Determine the equation of line between the points $(1,-1)$ and $(2,-2)$

$$\mathrm{V}_4(\mathrm{t})-1=\left(\frac{-2+1}{2-1}\right)(t+1)$$
$$\mathrm{V}_4(\mathrm{t})-1=-(t+1)$$
$$\mathrm{V}_4(\mathrm{t})=-t$$

But the Figure is lies between 1 and 2. Hence, multiply the expression with $\left[\mathrm{u}(\mathrm{t}-1)-\mathrm{u}(\mathrm{t}-2)\right]$

Hence,

$$\begin{aligned}\mathrm{V}_4(\mathrm{t})&=-t\left[\mathrm{u}(\mathrm{t}-1)-\mathrm{u}(\mathrm{t}-2)\right]\\&=-t\mathrm{u}(\mathrm{t}-1)+t\mathrm{u}(\mathrm{t}-2)\\&=-(t-1+1)\mathrm{u}(\mathrm{t}-1)+(t-2+2)\mathrm{u}(\mathrm{t}-2)\\&=-(t-1)\mathrm{u}(\mathrm{t}-1)-\mathrm{u}(\mathrm{t}-1)+(t-2)\mathrm{u}(\mathrm{t}-2)+2\mathrm{u}(\mathrm{t}-2)\\&=-r(\mathrm{t}-1)-\mathrm{u}(\mathrm{t}-1)+r(\mathrm{t}-2)+2\mathrm{u}(\mathrm{t}-2)\qquad\left[\text{Since }\mathrm{tu}(t)=r(t)\right]\end{aligned}$$

Therefore, the expression of $\mathrm{V}_4(\mathrm{t})$ is $\boxed{-r(\mathrm{t}-1)-\mathrm{u}(\mathrm{t}-1)+r(\mathrm{t}-2)+2\mathrm{u}(\mathrm{t}-2)}$.